

Agile Lab

Final Assessment Test

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Question:

Inventory Management System

InventoryService.java:

```
package com.inventory;
```

```
import java.util.HashMap;
```

```
import java.util.Map;
```

```
public class InventoryService {
```

```
    private Map<String, Integer> stockMap = new HashMap<>();
```

```
    public void addProduct(String productId, int initialStock) {
```

```
        stockMap.put(productId, initialStock);
```

```
    }
```

```
    public int updateStock(String productId, int quantityChange) {
```

```

        if (!stockMap.containsKey(productId)) {
            throw new IllegalArgumentException("Product ID not found");
        }

        int currentStock = stockMap.get(productId);
        int newStock = currentStock + quantityChange;

        if (newStock < 0) {
            throw new IllegalArgumentException("Insufficient stock level");
        }

        stockMap.put(productId, newStock);
        return newStock;
    }

    public int getStock(String productId) {
        return stockMap.getOrDefault(productId, 0);
    }
}

```

InventoryServiceTest.java:

```

package com.inventory;

import static org.junit.jupiter.api.Assertions.*;
import org.junit.jupiter.api.BeforeEach;
import org.junit.jupiter.api.Test;

public class InventoryServiceTest {

    private InventoryService inventoryService;

    @BeforeEach
    void setUp() {

```

```
inventoryService = new InventoryService();  
inventoryService.addProduct("PROD001", 50);  
}
```

@Test

```
void testStockIncrease() {  
    int updated = inventoryService.updateStock("PROD001", 10);  
    assertEquals(60, updated, "Stock should increase to 60");  
}
```

@Test

```
void testStockDecrease() {  
    int updated = inventoryService.updateStock("PROD001", -20);  
    assertEquals(30, updated, "Stock should decrease to 30");  
}
```

@Test

```
void testInsufficientStock() {  
    assertThrows(IllegalArgumentException.class, () -> {  
        inventoryService.updateStock("PROD001", -60);  
    }, "Should throw exception for negative stock result");  
}
```

@Test

```
void testInvalidProduct() {  
    assertThrows(IllegalArgumentException.class, () -> {  
        inventoryService.updateStock("INVALID_ID", 10);  
    });  
}  
}
```

mvn clean test:

```
Windows PowerShell

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[INFO] No tests to run.
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 58.548 s
[INFO] Finished at: 2026-04-13T10:35:29+05:30
[INFO] -----
```

mvn package:

```
Windows PowerShell

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[WARNING] JAR will be empty - no content was marked for inclusion!
[INFO] Building jar: C:\Users\NAVEEN BABU\OneDrive\Desktop\Inventory_management_system\src\target\inventory-management-system-1.0-SNAPSHOT.jar
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 50.102 s
[INFO] Finished at: 2026-04-13T10:36:37+05:30
[INFO] -----
```

GITHUB:

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